WeatherHub — ESP32 station hardware

A concrete shopping list for an ESP32 weather station that publishes into the WeatherHub backend over MQTT.

Phase 1 (this doc): tethered prototype. The board runs from a USB-C cable, no battery, indoor or sheltered placement. Goal is fastest path to "readings landing in the dashboard." ~$35–$50 from Amazon.

Phase 2 (notes at the end): outdoor / off-grid. Adds solar panel, 18650 cell, charging board, waterproof enclosure, Stevenson screen. Adds ~$40 and a weekend of mechanical assembly.

The backend ingests these metrics from each station, so the sensor set is chosen to fill them:

Metric	Field on `WeatherReading`	Sensor below
Temperature (°C)	`temperatureC`	BME280 / BME680
Humidity (%)	`humidityPct`	BME280 / BME680
Pressure (hPa)	`pressureHpa`	BME280 / BME680
Light (lux)	`lightLux`	BH1750
Air quality (AQI)	`airQualityValue`	BME680 (VOC) — Tier 2 swap
Rainfall (mm)	`rainfallMm`	Tipping-bucket rain gauge (optional)
Battery (V)	`batteryVoltage`	**null** in prototype (no battery)
Signal (dBm)	`signalRssi`	ESP32 WiFi (no extra HW)

The backend already tolerates `null` for every metric — fields you don't wire just don't get reported.

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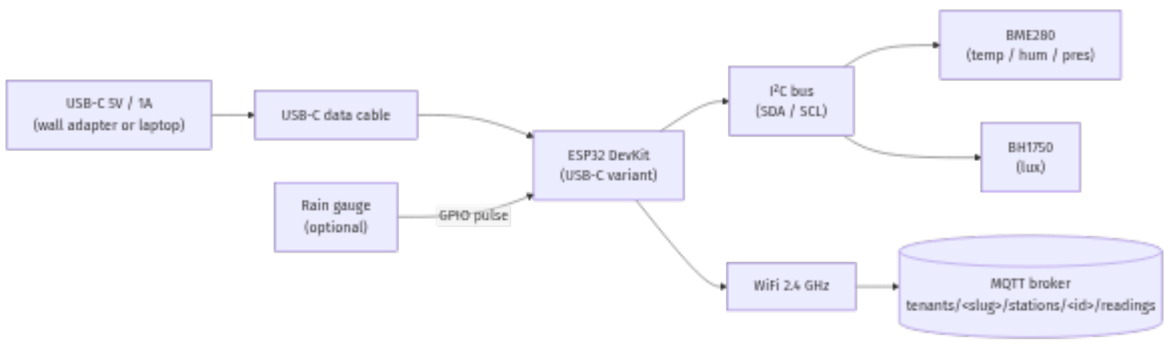
1. System topology — Phase 1 (USB-C tethered)



Total parts on the critical path: **ESP32 board, 2 I²C sensor modules, USB-C cable, a handful of jumper wires.** That's it.

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2. Bill of materials — Phase 1 prototype

Search terms are the ones that bring up the right part on Amazon US. Prices are typical 2026 ranges and will move with stock.

#	Part	Search term on Amazon	Qty	Est. $ each	Notes
1	ESP32 DevKit **with USB-C**	`ESP32-S3-DevKitC-1` or `HiLetgo ESP32 USB-C`	1	$10 – $15	Many "NodeMCU ESP32" boards on Amazon now ship with USB-C –
2	BME280 module (temp/hum/pres)	`BME280 I2C 3.3V module`	1	$5 – $8	Confirm BME**280** with humidity, not BMP**280**.
3	BH1750 module (lux)	`BH1750 GY-302 light sensor`	1	$4 – $6	I²C, accurate to ~65k lux.
4	USB-C data cable (5V/3A capable)	`USB-C charging cable data 3ft`	1	$5 – $8	**Data, not charge-only** — needed to flash + power.
5	USB-C power supply (or use a laptop)	`5V 2A USB-C wall charger`	1	$5 – $10	Skip if you'll power from a laptop / hub.
6	Dupont jumper wire kit (M-F, M-M)	`Dupont jumper wires 40-pin`	1	$5 – $8	4 wires per sensor (SDA, SCL, VCC, GND).
7	Solderless breadboard (400-tie)	`400 point solderless breadboard`	1	$4 – $7	Optional but makes wiring far cleaner than dangling Dupont.
	Subtotal			**~ $45**	

Optional add-ons (still Phase 1)

Part	Search term	Est. $	Why
Tipping-bucket	`Misol rain gauge` or		Adds `rainfallMm`. Only useful outdoors. Skip for indoor prototype.

rain gauge	`Sparkfun rain gauge`	$15 – $25	
BME680 (instead of BME280)	`BME680 I2C breakout`	$15 – $25	Adds `airQualityValue` (VOC index). Same wiring as BME280.
microSD card module + 32GB	`microSD card module SPI`	$8	Local buffering when WiFi drops. Nice-to-have, not needed for prototype.

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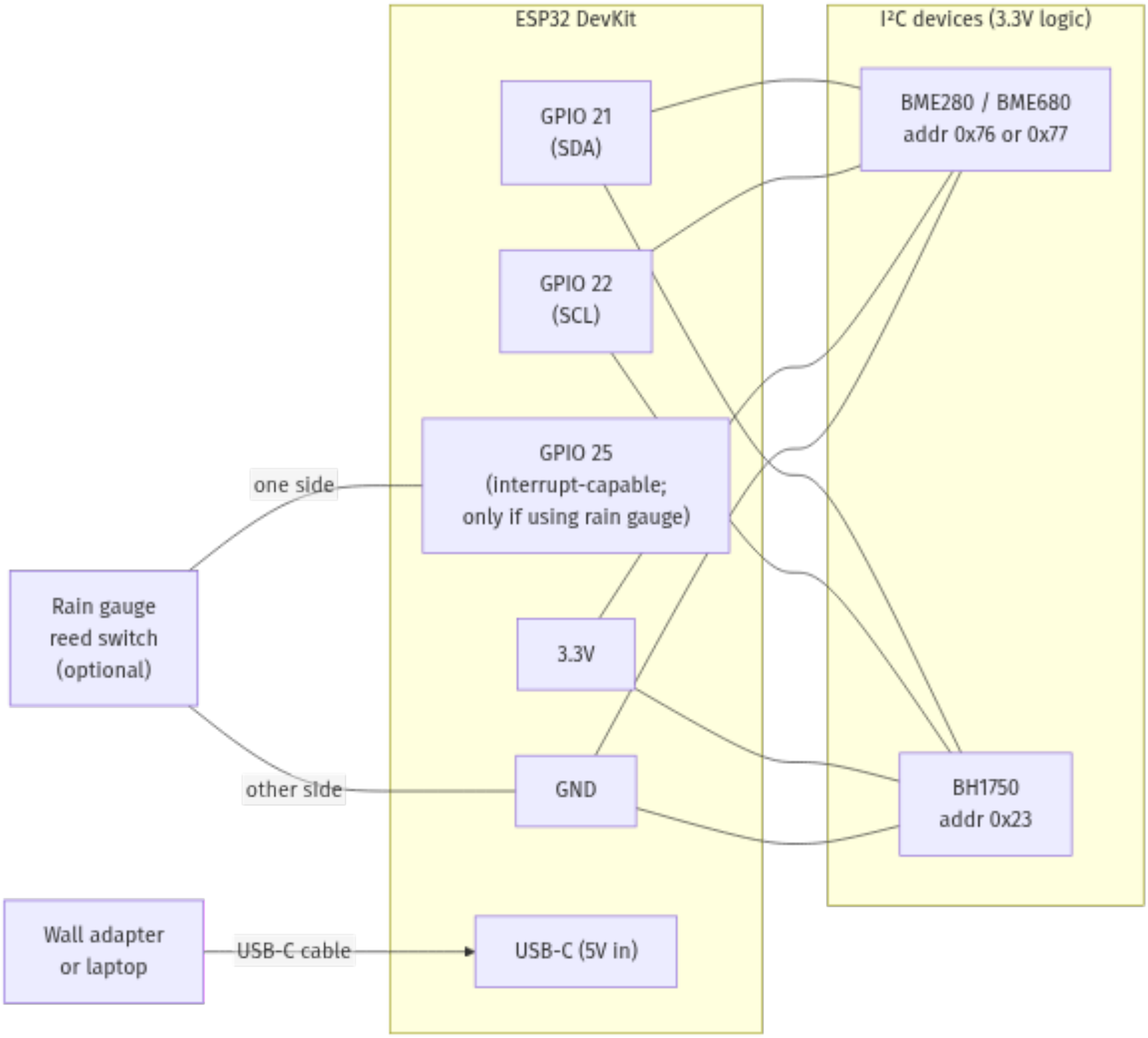
3. Wiring — GPIO pin assignment



Pin reference table:

ESP32 pin	Goes to	Notes
GPIO 21	I²C SDA on both BME280 + BH1750	Default I²C bus on most ESP32 boards

GPIO 22	I²C SCL on both	Default I²C bus on most ESP32 boards
3.3V	VCC on both I²C modules	**Do NOT use 5V on bare BME280** — it'll fry
GND	GND on every module	Common reference
GPIO 25	One side of rain-gauge reed (optional)	Interrupt-capable; skip if no rain gauge
USB-C	Wall adapter or laptop	Provides 5V + serial for flashing

I²C addresses don't collide:
- BME280: `0x76` (default) or `0x77` (SDO pulled high)
- BH1750: `0x23` (default) or `0x5C` (ADDR pulled high)
- BME680: `0x76` or `0x77` (same as BME280; swap parts, no rewire)

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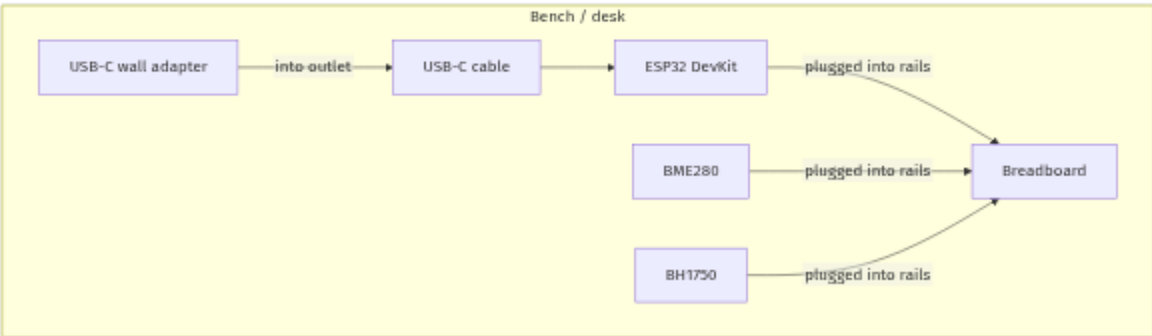
4. Physical assembly — Phase 1

Indoor / sheltered placement. No enclosure needed; the breadboard sits on a desk or shelf next to the wall outlet.



Placement tips for honest readings
- Keep the BME280 **away from the ESP32 body** — the regulator on the dev board self-heats and skews temperature by 1–3°C if mounted right next to it. A 10–15cm jumper wire is enough separation.
- Don't put the sensors on top of a laptop / radiator / sunlit windowsill. Even shaded indoor placement reads ~1°C higher than ambient.
- The BH1750 needs **direct line of sight to the light source** — pointing it at a wall is going to show 0 lux even in a bright room.

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5. Firmware checklist

Once the hardware is wired, the firmware needs to:
1. Connect to WiFi.
2. Read I²C sensors (BME280, BH1750).
3. Build the JSON payload:

```
{ "token": "<DEVICE_JWT>",
"reading": {
"recordedAt": "<ISO timestamp>",
"temperatureC": ..., "humidityPct": ..., "pressureHpa": ...,
"lightLux": ...,
"signalRssi": <wifi.RSSI()>
} }
```

Fields you don't measure (`rainfallMm`, `airQualityValue`, `batteryVoltage`) are omitted — the backend treats absent fields as null.

4. Publish to `tenants/<tenant-slug>/stations/<station-uuid>/readings` at QoS 1.

5. Sleep for the publish interval (5 min is the default the dashboard expects).

- **Tethered mode** can use `delay()` / `vTaskDelay()` — no need for deep sleep since power is unlimited.

Get the device JWT (and the exact topic) from:

```sh
docker compose exec backend node dist/database/scripts/device-provision.js \
--tenant=<slug> --station=<uuid>
```

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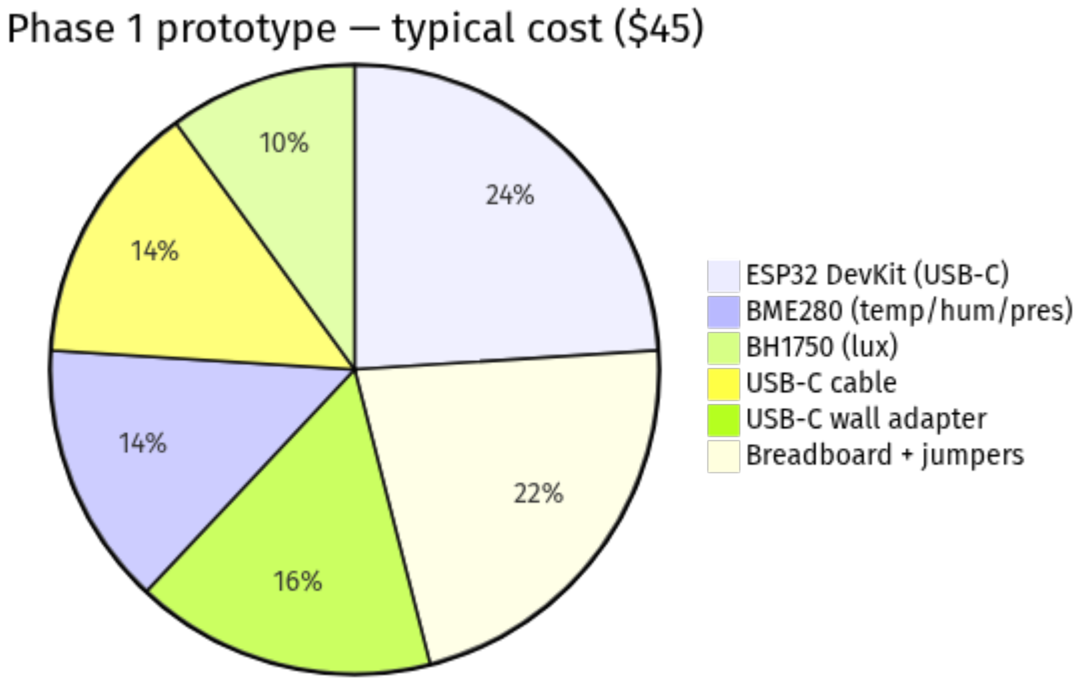
6. Cost roll-up



Add **~$20** for BME680 (air quality + VOC), or **~$25** for a rain gauge (only if outdoors).

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7. Pre-purchase sanity checklist

- [ ] ESP32 board has a **USB-C** connector, not micro-USB
- [ ] BME**280** (with humidity), not BME**P**280 — the spelling matters
- [ ] USB-C cable is rated for **data**, not charge-only — many bundled cables won't enumerate the COM port for flashing
- [ ] You have a USB-C wall adapter, OR you'll be powering the board off a laptop/hub during the prototype
- [ ] Sensor modules say **3.3V** in the listing — some breakouts are 5V-only and will fry

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8. Phase 2 preview — outdoor / battery / solar (not yet)

When you're ready to put a station outdoors:

Add	Est. $	Why
6V / 2W solar panel (ETFE)	$8 – $15	Charges the battery; sized for Tunis-latitude winter sun.
TP4056 charging module with protection chip	$1 – $3	DW01 IC visible on board. Without protection, the cell over-discharges.
18650 Li-ion cell (Samsung 25R / Panasonic NCR18650B)	$5 – $10	3000+ mAh real capacity. Avoid no-name "9999 mAh" listings.
18650 battery holder (solder tab)	$1 – $3	
Schottky diode 1N5819 / SS14	$0.50	Between solar `+` and TP4056 `IN+` so the cell can't drain back through the panel at
Voltage divider (2× 100kΩ resistors)	$0.10	Battery monitoring via ADC1 (GPIO 32). Reports `batteryVoltage` to the backend.
IP65/66 weatherproof enclosure (~200×150×100)	$10 – $18	Polycarbonate; clear lid if you want to read the LEDs through it.
Cable glands (PG7 / PG9)	$0.50	Waterproof cable entry — pointed downward always.

Stevenson screen (radiation shield)	$0 – $40	DIY from stacked PVC drain couplings (~$8) or pre-made ($30–40). Skipping this puts y
Tipping-bucket rain gauge	$15 – $25	Reed-switch contact closure per ~0.28 mm of rain.

Plus the firmware-side changes — deep sleep between publishes, ADC sampling for battery, ULP counter for rain pulses — which the Phase 1 firmware doesn't need.

A Phase 2 station lands around **$120 all-in** and runs for years off the solar panel without intervention. Phase 1 covers the same data plane on the bench so you can ship the software first, harden the hardware later.